THE MODULES ADMITTING THE STANDARD COKERNEL PRESENTATION FORM A WEAK SERRE SUBCATEGORY UNDER A HYPOTHESIS ON KERNELS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites ten external items (one homology criterion, one situation, several lemmas and an equation), all recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Criterion for a full subcategory to be a weak Serre subcategory. Stacks Project, Homology, Lemma homology-lemma-characterize-weak-serre-subcategory (tag 02N3). Reduces the verification to the kernel case.
- Closure of the standard presentation class under cokernels and extensions. Stacks Project, Modules on Sites, Lemmas lemma-cokernel-map-towards-constructibles (tag 09BC) and lemma-extension-towards-constructibles (tag 09BD). Supplies the two closure results that the reduction uses.

Source reference. Original reference 0937: sources/stacks-chow/tags.txt, source lines 11505–11505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0938: sources/stacks-chow/tags.txt, source lines 11506–11506. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0939: sources/stacks-chow/tags.txt, source lines 11507–11507. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 093A: sources/stacks-chow/tags.txt, source lines 11508–11508. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 093D: sources/stacks-chow/tags.txt, source lines 11511–11511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0754: sources/stacks-chow/tags.txt, source lines 9118–9118. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 093E: sources/stacks-chow/tags.txt, source lines 11512–11512. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:sites-modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 093G: sources/stacks-chow/tags.txt, source lines 11514–11514. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0934: sources/stacks-chow/tags.txt, source lines 11502–11502. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 093F: sources/stacks-chow/tags.txt, source lines 11513–11513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0937 assume (0938), (0939), and (093A) hold. Let $\mathcal{O}$ be a sheaf of rings. Let $\mathcal{A} \subset Mod(\mathcal{O})$ be the full subcategory of modules isomorphic to a cokernel as in (093D). If the kernel of every map of $\mathcal{O}$-modules of the form*

$$\bigoplus\nolimits_{j=1,\ldots,m} j_{V_j!}\mathcal{O}_{V_j} \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} j_{U_i!}\mathcal{O}_{U_i}$$

with U_i and V_j in $\mathcal{B}$, is in $\mathcal{A}$, then $\mathcal{A}$ is weak Serre subcategory of $Mod(\mathcal{O})$.

Proof. We will use the criterion of Homology, Lemma 0754. By the results of Lemmas 093E and 093G it suffices to see that the kernel of a map $\mathcal{F} \to \mathcal{G}$ between objects of $\mathcal{A}$ is in $\mathcal{A}$. To prove this choose presentations

$$\bigoplus A_{V_j} \to \bigoplus A_{U_i} \to \mathcal{F} \to 0 \quad \text{and} \quad \bigoplus A_{T_j} \to \bigoplus A_{W_i} \to \mathcal{G} \to 0$$

In this proof the direct sums are always finite, and we write $A_U = j_{U!}\mathcal{O}_U$ for $U \in \mathcal{B}$. Using Lemmas 0934 and 093F and arguing as in the proof of Lemma 093G we may assume that the map $\mathcal{F} \to \mathcal{G}$ lifts to a map of presentations

$$\begin{array}{ccccccc} \bigoplus A_{V_j} & \longrightarrow & \bigoplus A_{U_i} & \longrightarrow & \mathcal{F} & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ \bigoplus A_{T_j} & \longrightarrow & \bigoplus A_{W_i} & \longrightarrow & \mathcal{G} & \longrightarrow & 0 \end{array}$$

Then we see that

$$\mathrm{Ker}(\mathcal{F} \to \mathcal{G}) = \mathrm{Coker}\left(\bigoplus A_{V_j} \to \mathrm{Ker}\left(\bigoplus A_{T_j} \oplus \bigoplus A_{U_i} \to \bigoplus A_{W_i}\right)\right)$$

and the lemma follows from the assumption and Lemma 093E. □